Bank Multi-Carrier), UFMC (Universal Filtered MultiCarrier), GFDM (Generalised Frequency Division Multiplexing), f-OFDM (Filtered - Orthogonal frequency-division multiplexing). Besides these, there are other waveforms under consideration as well.

Currently, no single waveform can support the wish list 5G is vying for and it is to be believed that in the end, a combination of wavelengths or an adaptive wavelength will be able to provide the required results. This also means, that scientists will have a to develop a whole new receiving and transmission architecture to accommodate all the features of the 5G standard. However, at the moment it's still under development.

While companies like Ericsson, Nokia and Huawei are hosting demos of what they have achieved so far, it is still the early days for the technology. As years go by and 5G comes to life, it will fall to service providers to push and adopt the technology further. This will require new infrastructure at every stage. After all, 5G mobile communications are supposed to handle almost 1,000 times more traffic, 10Gbps of speed and super low latency of under 1ms.

How is the research going?

It all comes down to the research of the various aspects which will shape the mobile broadband of the future. As discussed, higher frequency bands are likely to be used and to counter the problem of smaller coverage, a new technology called 'beam-forming' has been widely accepted. The concept behind it is quite simple. Instead of broadcasting the signal in a wide area, concentrate the signal directly to the target device. Some new Wi-Fi routers use this technology and we might see an evolved variant of the technology in the upcoming 5G standard. This technology proposes to focus radio interface into beams which will make them usable over long distances. These beams must be directed towards the end user and since the service is required to be mobile and available to many, the corresponding combination of hardware and software has to track and stay connected to each individual in the vicinity. This is quite a task, keeping in mind that each cell can only support a certain number of beams at a time and that too, without dropping the quality of the network.

Fact

A cell is simply a geographical area covered by a cellular telephone transmitter. It can vary from 1KM to 30KM in diameter

The other technology which is being worked upon is MIMO (Multi-Input, Multi-Output). This technology requires the devices to be fitted with an array of antennas via which multiple radio connections can be made between the device and the cell tower. However, high-order MIMO has an issue, it causes radio interference. Hence, a new method or technology is required to transfer data via MIMO. This needs a new kind of radio network which can adjust its beam to take into account the specific orientation of the antenna at any given time. The idea of 3D MIMO is also being researched, which will allow beam-forming in 3D.

To counter the massive demands for data, an easier solution of dense networks is being considered. This means reducing the cell size. By reducing the cell size, and lowering the range, spectral efficiency can be improved as the capacity increases. However, this will also mean more number of base stations will be required, again increasing the cost incurred on the operator. So, research is being done to improve that.

Currently, the existing cellular communication uses two full-duplex form, FDD (frequency Division Duplex) and TDD (Time Division Duplex), which essentially means using two different frequencies to transmit and receive. In FDD, the network uses different frequencies to receive and transmit and in TDD, the transmission and reception is done at different times. However, both these methods are not completely efficient. So, a new full duplex method is in the works which will allow simultaneous transmission and reception on the same channel at the same time.

Work is also going on Multi-Radio Access technologies which will essentially allow devices to talk to the various radio interfaces and access points at the same time. It is likely to involve Wi-Fi, higher frequencies and currently used frequencies into the mix.

As mentioned earlier, 5G is likely to work on a higher spectrum using millimeter waves but our current research is inadequate to make the system work. Beyond this, there are other goals such as "1ms latency" which needs to be achieved and for that, the technology needs to evolve. There are multiple research teams around the world trying to solve that issue and are going forward towards different research paths. We just hope that by 2020, they actually make it a reality.